PADHAKU
Curiosity · Class 6 Science



CHAPTER 12

Beyond Earth

NCERT Exercise Solutions

Activity 12.1: Let us draw	1 Q
12.1 Stars and Constellations	2 Q
12.2 Night Sky Watching	4 Q
12.3 Our Solar System	8 Q
12.4 The Milky Way Galaxy · 12.5 The Universe	2 Q
Let us enhance our learning	12 Q
Learning further	5 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

Independent study aid by Beginers Point. Not an official NCERT publication.

Activity 12.1: Let us draw

12.1 **Look at Fig. 12.2 (bright stars in one part of the night sky). Imagine a pattern formed by a group of stars, join the stars with lines and name the animal or object it looks like.**

Compare your patterns with your friends'. Are the patterns same or different?

💡 **SAMPLE ANSWER**

- Example: four stars in a box with three stars in a line look like a **cot** or a **ladle (spoon)**; another group may look like a **kite** or a **bird**. (check the picture in your book)
- Everyone's patterns, names and stories are **different**, because the patterns are only imaginary lines that each person draws in their own way.
- Our ancestors did the same: they named star patterns after animals, objects and characters in stories.

12.1 Stars and Constellations

? **Do we find patterns among the stars just for fun or is there some use of these patterns?**

✔ **ANSWER**

They are useful too.

- Patterns help us **recognise stars** in the sky.

- In olden times, before the magnetic compass, sailors and travellers used them to **find directions** at sea or on land. It is still a backup method in emergencies.

? How can we identify some of these constellations in the night sky?

✔ ANSWER

- Learn how the constellation looks, using pictures like Fig. 12.3 and Fig. 12.4.
- Use a sky map app (Sky Map, Stellarium) to know when and in which part of the sky it will be seen.
- Watch from a dark, open place on a clear night, with an adult.

⭐ 12.2 Night Sky Watching

? Can we try to find any constellation or star of our choice on any night from our location?

✔ ANSWER

No. Not all stars and constellations are visible from all places on Earth and on all nights of the year.

For example, the Pole Star cannot be seen from the southern hemisphere, and Orion is best seen in India from December to April. A sky map app tells us when and where to look.

12.2 Activity 12.2: Let us try to locate. Find the Big Dipper and then the Pole Star.

✔ ANSWER

- Around 9 p.m. in summer, look at the **northern** part of the sky and find the Big Dipper (Saptaṛiṣhi): 4 stars like a cup and 3 like a handle.
- Join the two stars at the end of the cup with an imaginary line and extend it towards the north.
- At about **five times** the distance between these two stars, we find a star that is not very bright. This is the **Pole Star**.

Conclusion: The Big Dipper helps us find the Pole Star, which shows the North direction.

12.3 Activity 12.3: Let us try to identify. Find Orion and the star Sirius.

✔ ANSWER

- From December to April, after sunset, look for **three bright stars in a short straight line**. They form the belt of the hunter in Orion.
- Imagine a straight line through these three stars and look along it towards the **east**. It leads to a very bright star: **Sirius**.

Conclusion: Orion's belt helps us find Sirius, the brightest star in the night sky.

? **Which star is closest to us?**

✔ ANSWER

The **Sun** is the star closest to us. After the Sun, the nearest star is **Proxima Centauri** (about 269000 au away).

⭐ 12.3 Our Solar System

? **Our Sun is also a star. So how is it that the Sun appears big and lights up the sky while the other stars look like bright dots in the night sky and are not even visible during the day time?**

✔ ANSWER

- The Sun is **much closer** to us than other stars, so it looks big and very bright.
- Other stars are **very far away**, so they look like points, even though some are much bigger than the Sun.
- In the day, the **extreme brightness of the Sun** makes it impossible to see the other stars.

? **Are stars the only objects in the sky? Or, are there more objects which we may not have noticed?**

✔ ANSWER

No. There are many more objects: **planets, moons, asteroids and comets**. Together with the Sun they form our **Solar System**.

? Can we identify a planet the same way we identified the constellations?

✔ ANSWER

Partly. Venus is easy to find as it is very bright, and Mercury, Mars, Jupiter and Saturn can be seen with the naked eye. But they look like shining dots, just like stars.

We tell them apart because **stars twinkle a lot, planets do not**. A sky map app helps find their position.

12.4 Activity 12.4: Let us try to identify Venus.

✔ ANSWER

- At **dawn**: look near the **East** before sunrise.
- At **dusk**: look near the **West** after sunset.

Observation: Venus is a very bright, steady (not much twinkling) point of light. After the Sun and the Moon, it is the brightest object in the sky. It is called the Morning Star or Evening Star, but it is a planet.

? How can we see the planets which are not visible to the naked eye?

✔ ANSWER

By using a pair of **binoculars** or a **telescope**. A telescope shows objects brighter and larger and also shows dim objects that we cannot see with naked eyes.

? **We know that planets move around the Sun. Are there any objects that move around planets?**

✔ ANSWER

Yes. Objects that move around planets are called **satellites**. Moons are natural satellites, e.g. Earth has one Moon, Mars has two moons.

? **We have learnt about the objects which form the Solar System. Which are these objects?**

✔ ANSWER

The **Sun, eight planets, their moons**, and many smaller objects including **asteroids and comets** together form the Solar System.

? **What is beyond the Solar System?**

✔ ANSWER

Beyond the Solar System are other **stars**. The Sun and all these stars (millions to billions) form our galaxy, the **Milky Way Galaxy**.

12.4 The Milky Way Galaxy · 12.5 The Universe

? **What is beyond the Milky Way Galaxy?**

✔ ANSWER

There are **many other galaxies** in outer space beyond the Milky Way. Together they are part of the **Universe**.

? **Is there life anywhere else in the Universe?**

✔ ANSWER

We **do not know yet**. Scientists are searching, mostly on **exoplanets** (planets revolving around other stars), but no evidence of life has been found so far. The search is continuing.

Let us enhance our learning

1 **Match the column:**

Column I	Column II
(i) Satellite of Earth	(a) Orion
(ii) Red planet	(b) Venus
(iii) Constellation	(c) Mars
(iv) Planet which is commonly called an evening star	(d) Moon

✔ ANSWER

- (i) Satellite of Earth → **(d) Moon**
- (ii) Red planet → **(c) Mars**
- (iii) Constellation → **(a) Orion**
- (iv) Evening star → **(b) Venus**

2 (i) **Solve the following riddle.**

My first alphabet is in MAN but not in CAN

My second alphabet is in ACE and also in FAN

My third alphabet is in RAT and not in CAT

My fourth alphabet is in SUN but not in FUN

I am a planet that moves around the Sun.

✔ ANSWER

- MAN but not CAN → **M**
- in ACE and FAN → **A**
- RAT but not CAT → **R**
- SUN but not FUN → **S**

Answer: **MARS**

2 (ii) **Make two similar riddles by yourself.**

💡 SAMPLE ANSWER

Riddle 1: My first letter is in MOP but not in TOP. My second is in BOX and also in DOG. My third is in OWN but not in OWL. My fourth is in ROSE but not in NOSE. I go around the Earth. → **MOON**

Riddle 2: My first letter is in SIT but not in BIT. My second is in TUB and also in CUP. My third is in PIN but not in PIG. I am the star closest to us. → **SUN**

3 **Which of the following is not a member of our Solar System?**

(i) Sirius (ii) Comets (iii) Asteroids (iv) Pluto

✔ ANSWER

(i) Sirius. It is a star far away, in the constellation Canis Major. Comets, asteroids and Pluto revolve around the Sun.

4 **Which of the following is not a planet of the Sun?**

(i) Jupiter (ii) Pluto (iii) Neptune (iv) Saturn

✔ ANSWER

(ii) Pluto. Since 2006 (IAU definition), Pluto is called a **dwarf planet**.

5 **Which is the brighter star, the Pole Star or Sirius?**

✔ ANSWER

Sirius is brighter. It is the brightest star in the night sky, while the Pole Star is not very bright.

6 **An artist's representation of the Solar System is given in Fig. 12.12. Is the order of the planets correct? If not, write the correct order in the boxes in the figure.**

💡 **SAMPLE ANSWER**

No, the order in the figure is not correct (check the picture in your book).

Correct order from the Sun:

Mercury → Venus → Earth → Mars → Jupiter → Saturn → Uranus → Neptune

7 **A portion of night sky with stars is shown in Fig. 12.13. Identify the groups of stars that form the Big Dipper and the Little Dipper, draw lines to connect them and label them. Also identify and label the Pole Star.**

💡 **SAMPLE ANSWER**

- **Big Dipper:** find 7 bright stars: 4 forming a cup (a rough rectangle) and 3 in a curved line forming the handle. Join and label them.
- **Pole Star:** extend the line through the two end stars of the cup about 5 times their gap; the not-very-bright star there is the Pole Star.
- **Little Dipper:** the Pole Star is the end of its handle; join it with the other 6 fainter stars (handle + small cup) and label it.

(check the picture in your book)

8 **A portion of the night sky is shown in Fig. 12.14. Draw lines to connect the stars for Orion and label the star Sirius.**

💡 SAMPLE ANSWER

- Find the **three stars in a short straight line** (Orion's belt) in the middle.
- Join them with the stars above and below to make the hunter's shape (as in Fig. 12.3).
- Extend the belt line towards the east; the very bright star it reaches is **Sirius**. Label it.

(check the picture in your book)

9 **From Earth, you can see stars fading away at dawn and appearing at dusk. During the day we do not see the stars. Explain why.**

✔ ANSWER

- Stars are in the sky during the day also.
- But the **Sun is extremely bright** in the day, so its light hides the faint light of the far-away stars.
- At dawn, sunlight increases, so stars fade away; at dusk, sunlight decreases, so stars appear again.

10 **During a clear night, try to observe the Big Dipper 3–4 times at an interval of 2 to 3 hours. Also try to locate the Pole Star each time. Does the Big Dipper appear to move?**

Draw a rough sketch to illustrate this, mentioning the time in each case.

💡 SAMPLE ANSWER

Yes, the Big Dipper appears to move. It slowly turns in a circle around the Pole Star (anticlockwise).

The **Pole Star** stays nearly at the same place in the North each time.

Sketch: mark the Pole Star in the middle and draw the Big Dipper at each time, e.g. 8 p.m., 10 p.m., 12 midnight, 2 a.m., each turned a little further around the Pole Star.

11 **Think about the night sky and write a poem or a story on it.**

💡 SAMPLE ANSWER

Poem: The Night Sky

Little stars in a sky so deep,
Shining bright while the world's asleep.
The Pole Star shows us where North lies,
The Big Dipper turns across the skies.
Far from city lights we see,
A Milky Way of mystery!

📘 Learning further

1 **Try to find out the names of planets in your local language. Also, find out the stories**

associated with stars and constellations in your region. Present these stories in a pictorial form.

💡 SAMPLE ANSWER

Example (Hindi names): Mercury – Budh, Venus – Shukra, Earth – Prithvi, Mars – Mangal, Jupiter – Brihaspati/Guru, Saturn – Shani.

Example story: tribes of Central India see four stars of the Big Dipper as a **grandmother's cot** and the other three as **thieves** stealing it. Draw this as a picture story.

2 **If there is a planetarium or a science museum nearby, visit it, especially when a night sky observation session is scheduled.**

💡 SAMPLE ANSWER

Sample: At a night sky session we can see the Moon, planets and stars through a telescope. In the daytime, we can see models, pictures and sky shows. Write down what you saw.

3 **Find out if the increasing light pollution is causing problems for humans, wildlife and the environment. Write an action that you would take at the personal level to control light pollution.**

💡 SAMPLE ANSWER

- **Humans:** bright light at night can disturb sleep and health; we cannot see stars.
- **Wildlife:** birds, insects and turtles that move at night get confused by artificial light.
- **Environment:** wasted electricity.

My action: switch off lights not in use, use covered lights that shine downwards, and avoid extra decorative lights.

4 **Find out the weather conditions due to which the Indian Institute of Astrophysics (IIA) found Hanle, Ladakh a suitable place to set up an observatory.**

💡 SAMPLE ANSWER

- Very **dry** air and very few **cloudy** days (clear skies on most nights).
- Very **high altitude**, so less air above to blur the view.
- **Dark skies** with almost no light pollution (now the Hanle Dark Sky Reserve).

5 **If you enjoy doing embroidery, try to embroider on a dark coloured cloth the constellations that you have seen. Otherwise, depict constellations in other ways using art and craft ideas.**

💡 SAMPLE ANSWER

> Sample: Embroider Orion and the Big Dipper with white or yellow thread on a dark blue cloth. Other ideas: stick shining stickers or make holes in a black chart paper and hold it against light.

🎉 **Chapter 12 done! Revise the answers once more tomorrow.**